॥श्रीगणेशाय नमः॥

Percentages

$50\% = 1/2$ $30\% = 3/10$ $33.33\% = 1/3$ $1/4 = 25\%$ $1/5 = 20\%$ $1/6 = 16.66\%$

$1/7 = 14.2857\%$ $1/8 = 12.5\%$ $1/9 = \frac{1}{3} \cdot (33.33\%) = 11.11\%$

• $X\%$ of $Y = Y\%$ of X

Q. 50% of $P = 25\%$ of Q. $P = x\%$ of Q. Find x.

→ $\frac{1}{2}P = \frac{1}{4}Q$

$P = \frac{1}{2}Q$

$x = 50$

Q. 20% of $(p+q) = 50\%$ of $(p-q)$. $p : q = ?$

→ $2(p+q) = 5(p-q)$

$\frac{p+q}{p-q} = \frac{5}{2}$

$\frac{p}{q} = \frac{7}{3}$

Q. $20\%(a+b) = 50\%\ b$. $\frac{2a-b}{2a+b} = ?$

→ $2(a+b) = 5b$

$2a = 3b$

$\frac{2a-b}{2a+b} = \frac{2b}{4b} = 1/2$

Q. If x is 20% less than y, then $\frac{y-x}{y}$ and $\frac{x}{x-y}$ are ___ and ___.

→ $x = \frac{8}{10}y = \frac{4}{5}y$

$5x = 4y$

$\frac{y-x}{y} = \frac{y-\frac{4}{5}y}{y} = \frac{1}{5}$ $\frac{x}{x-y} = \frac{x}{x-\frac{5}{4}x} = -4$

Q. $60\%\ A = 30\%\ B$. $B = 40\%\ C$. $C = x\%\ A$. $x = ?$

→ $\frac{60}{100}A = \frac{30}{100} \times \frac{40}{100} \times \frac{x}{100} A$

$1 = \frac{20}{100 \cdot 100}x$

$x = 500$

X is what % of Y?

Q. 18 is what % of 9? → 200

Q. ½ is what % of ⅓ → $\frac{1}{2} = \frac{x}{100} \times \frac{1}{3} \Rightarrow 150$

What % of X is Y?

$\left(\frac{y}{x} \times 100\right)$

X is what % more/less than Y?

$(x-y) = \frac{?}{100} * y$

$\Rightarrow \frac{x-y}{y} \times 100$

Q. 2 #s are $12\frac{1}{2}\%$ and 25% more than a 3rd #. The 1st # is what % of 2nd?

→ $x = 1.125z \Rightarrow x = 1.125 \times \frac{y}{1.25}$

$y = 1.25z$

$x = (0.9)y$

90%

Q. #s are less than a 3rd # by 10% and 37% resp. The % by which 2nd is less than 1st?

→ $x = 0.7z$ $\quad x = \frac{0.7}{0.63} \times \frac{y}{9}$

$y = 0.63z$

$y = \frac{9}{10}x$

Q. Kishan spends 30% of sal on food, donates 3%. That's ₹2310. What's sal?

→ $\frac{33}{100}$ sal = 2310

sal = 7000

Practice

Q. 2 persons contested in an election. The winner secured 57% of the total votes and won by a majority of 42,000 votes. Find # of votes?

→ winner → 57%
loser → 43%
diff → 14%

$\frac{14}{100} x = 42000$

$x = 3,00,000$

Q. A candidate secured 40% votes but is defeated by the only other candidate by 298 votes.

→ 60% − 40% = 20%

$\frac{298}{149} = \frac{20}{10} x$ $\quad x = 1490$

Q. 2 candidates. One wins with 72%. If total = 8200, won by how much?

→ 72% − 28% = 44% $\quad \frac{44}{100} \times 8200 = 3608$.

Q. 3 candidates. 1st got 40%, 2nd → 36%, Total votes = 36000, find # of votes got by 3rd.

→ $\frac{24}{100} \times 36000 = 8640$

Q. 2 candidates. 10% votes are invalid. winner gets 70% of valid votes and defeats the loser by 1800. Total # ?

→ $\left(\frac{70}{100} \times \frac{90}{100} - \frac{30}{100} \times \frac{90}{100}\right) x = 1800$

$\left(\frac{40}{100}\right)\left(\frac{90}{100}\right) x = 1800$

$x = \frac{10000}{2} = 5000$

Q. 8% voters didn't vote. 2 candidates. winner → 48% votes (of total). won by 1100. # of total voters?

→ 48 − 44 = 4

$\frac{4}{100} x = 1100$

Q. A candidate got 55% of total valid votes. 2% of total → invalid. Total # of voters → 104000, # of valid votes in favor of candidate?

→ $0.55 \times 0.98 \times 104000 = 56056$

Q. In an exam → 1000 boys, 800 g. 60% b, 50% g passed. Find % of candidates failed

→ 600 B, 400 g = 1000 passed ⇒ 800 failed.

$$800 = \frac{x}{100} \times 1800$$

$$x = 44.44$$

Q. Exam. A candidate must secure 40% to pass. One who gets 220 fails by 20. Max marks?

→ $240 = \frac{40}{100} x \quad \Rightarrow x = 600$

Q. 36% to pass. A student got 113. Failed by 85.

→ $(113+85) = \frac{36}{100} x \quad \Rightarrow x = 550$

Q. 33% to pass. Got 25%. Failed by 40

→ $8\% \; x = 40 \quad \Rightarrow x = 500$

Q. Gets 20%. Fails by 5.
Another gets 30%, gets 20 marks more than passing.

→ Say passing is at x %.

$\frac{(x-20)}{100}$ total = 5

$\frac{(30-x)}{100}$ total = 20

$$\frac{x-20}{30-x} = \frac{5}{20} = \frac{1}{4}$$

$$4x - 80 = 30 - x$$

$$5x = 110$$

$$x = 22$$

Q. 3 subjects, 100 marks each. A student scores 60%, 80% in 1st & 2nd, 70% in aggregate.

→ 1st 60, 2nd 80, 3rd x

$$140 + x = \frac{70}{100} \times 300 = 210 \quad \Rightarrow x = 70$$

Q. A # reduced by 20%. becomes 228. How much inc to get 313.5?

→ $80\% \; x = 228$

$x = 285$

diff = 28.5

$$28.5 = \frac{x}{100} \times 285$$

$$x = \frac{2850}{285} = 10$$

Q. Country popⁿ: 1400 million. 70% own mobile phones. Among those, 294 m access the internet. Among those, ½ buy goods online. What is their %?

→ $\frac{1}{2} \times 294 = 147$

$$147 \text{ m} = \frac{x}{100} \times 1400 \text{ m}$$

$$x = \frac{14700}{1400} = 10.5$$

Q. Fiscal deficit = 4% of GDP in 2015.
5% in 2016.
GDP inc. = 10%. % inc. in actual Fiscal deficit?

→ Let $FD_1 = x$. $x = \frac{4}{100}$ GDP

$25x = GDP$

new GDP = 1.1 GDP = $1.1 \times 25 \times x = 27.5x$

$FD_2 = \frac{5}{100} \times 27.5x = \frac{137.5x}{100}$

Inc. in FD is 37.5%

Q. price of sugar ↑ by 20%, how much % of consumption should be decreased s.t. exp. remains same?

→ Exp. = sugar × amount

$$\frac{120}{100}\ \text{sugar} \times \frac{x}{100}\ \text{amount}$$

$$120x = 10000$$

$$x = 83.333$$

∴ Dec. by 16.66 %

Q. Party. 60% invited guests are male. 80% of guests attend. All the invited females attended. 40% female. Ratio of m : f attended?

→

100 inv.

60 m	40 f
x	40

$x + 40 = 80$

$x = 40$

1 : 1

Q. Designer uses 4 diff. colored marbles. Cost same for each.
Table shows % of marbles of each color used in current design.

Blue	40 %
Black	25 %
Red	20 %
yellow	15 %

cost of each marble is inc. by 25%.
The designer decided to reduced equal #s of marbles of each color to keep total cost unchanged.
% of blue marbles in new design?

→ cost = 100 × 1

$40 - x$
$25 - x$
$20 - x$
$15 - x$
$100 - 4x$

$$100 \times 1 = (100 - 4x) \times 1.25$$

$$100 - 4x = \frac{100}{1.25}$$

$$4x = 100 - \frac{100}{1.25}$$

$$x = \frac{20}{4} = 5$$

Blue → 35
→ 20
→ 15
→ 10
80

$$35 = \frac{y}{100} \times 80$$

$$y = \frac{350}{8} = 43.75$$

(x % of y) + (y % of x) is ——
a) 2 % of (xy/100) b) xy % of 100
c) 2 % of xy d) 100 % of xy

→ $\frac{xy}{100} + \frac{xy}{100} = \frac{2}{100} xy$ = 2% of xy

Q. Apples and oranges. 15% are unripe fruits. Of those, 45% are apples.
Of the ripe ones, 66% are oranges.
Pile has total 5692000 fruits, how many of them are apples.

→ $0.15 \times 0.45 \times 5692000$

$= 384210$ ← unripe apples

$\frac{34}{100} \times \frac{85}{100} \times 5692000 = 1644988$ ← ripe apples

Total apples $= 2029198$

Q. GDP grew 7% (2012→13). For international comparison, it is compared in USD based on the exchange rate.
(₹)

During the period, exchange rate for USD inc. ₹50/usd → ₹60/$.
India's GDP in USD grew by ___ %.
a) +5% b) −13% c) −20% d) −11%

→ $x \longrightarrow 1.07x$

Internat'l: $\frac{x}{50} \longrightarrow \frac{1.07x}{60}$

$$\frac{1.07x}{60} = \frac{y}{100} \times \frac{x}{50}$$

$$y = \frac{107 \times 50}{60} = 89.166$$

−11%

Total questions = 30
Q. Manisha did more than 50% right answers than she did wrong answers.
Each ans was right/wrong.
How many questions did she get right?

→ wrong x → right $1.5x$

$2.5x = 30$
$x = 12$
$30 - x = 18$

Q. 2 identical cakes. Ram gets 37.5% and 56.25% of them.
Shyam orders a full cake. Find the value y: $\frac{\text{diff. in amount of cakes Ram \& Shyam have}}{\text{weight of 1 cake}}$

→ say wt. $= w$

Ram gets $\frac{93.75w}{100}$

Shyam gets $\frac{100}{100}w$

$y = \frac{6.25}{100}\frac{w}{w} = 0.0625$

Q. Price 24% ↑ consumption 25% ↑. Expenditure ↑?
→ $1.24 \times 1.25 = 1.55$

55%